

Phillip Williams

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EDUCATION AND HONORS

Duke University, Pratt School of Engineering

B.S.E. in Electrical & Computer Engineering, B.S. in Computer Science

- ❖ Balanced two jobs on top of my degree coursework simultaneously in order to finance my college education.
- ❖ Received **A. James Clark Scholarship for Engineering and Entrepreneurship**, **Sky Polega Scholarship**, and **TAPIA 2025 Conference Scholarship**.

PROFESSIONAL EXPERIENCE

AI Software Engineer, 05/2025 to 08/2025

Tribe.World – Bootstrapped startup developing a trust-first, skills-based hiring platform

- ❖ Architected AI pipelines using **LangGraph** and **Ollama** to automate skill extraction from unstructured resumes and job descriptions, achieving **more than 7x improvement** in extraction count over Workday.
- ❖ Designed and delivered a cross-platform application (**Firestore**, **Flutter**). Integrated skills extraction and matching pipelines, ranked candidate reports (categorizing skills as met/unmet and required/desired), user authentication/registration flows, and profile review functionality.
- ❖ Developed a proprietary matching algorithm that outperformed BM25 and semantic matching by identifying indirect skill correlations that traditional methods miss, increasing match specificity for hiring managers.
- ❖ Created a data visualization tool in JavaScript to render learned data structures as human-editable interfaces, removing the need for manual code updates and allowing audit from industry professionals.
- ❖ Designed and built a lightweight tool (**Typescript**, **Next.js**, **Postgres**, **Ollama**) to generate a dataset of resume and job description documents as a supplement to a dataset of human-written documents.
- ❖ Used **Cursor** and **AI agents** on parallel workbranches to accelerate development.

Software Engineer / VFX Producer, 01/2022 to present

Prosper XR – Immersive content creation studio startup with specialties in robotics and VR

- ❖ Produced [personalized college graduation experiences in virtual reality](#) scaling to over **42,000 users**, resulting in **4.5x** more views for the primary client.
- ❖ Engineered a custom automation pipeline in **TypeScript** for Adobe After Effects, cutting render time by ~6,000 hours annually (**\$180k/year** cost savings).
- ❖ Wrote and maintained **63 pages of documentation** detailing the pipeline and editing workflow for future use.
- ❖ Managed our registration database, **communicated directly with clients and end users**, and resolved support tickets to ensure client satisfaction.

XR Software Developer / VR Lab Guide, 09/2022 to 12/2025

Duke University Multimedia Project Studio – VR for faculty research, game development, 3D modeling

- ❖ Developed a **Unity-Arduino** serial interface in **C#** and **C** to automate stimuli and feedback in an avian research project in Duke's Nowicki Biology Lab.
- ❖ Simplified our lab manager's workflow for collecting this experiment's data by building a GUI in **Java Swing**.
- ❖ Directed graduate and undergraduate students in the use of VR equipment and VR application development (**Unity**, **C#** and **Maya**).

PROJECTS - these (and more) available @ github.com/pcaw627

Real-Time Digital Equalizer for Electric Guitar, Pipelined CPU on FPGA

- ❖ Architected and implemented RTL design for a pipelined CPU in **Verilog** with Wallace Tree Multiplier, pipelining, bypass, and hazard logic.
- ❖ Designed a real-time audio equalizer on top of this CPU by implementing a DSP pipeline using Fast Fourier Transform (FFT) and Inverse FFT in **Verilog**, also leveraging the CPU's ISA in **Assembly**.
- ❖ Deployed on an Artix-7 **FPGA**, and built to interface with an electric guitar and amp via ADC/DAC.
- ❖ Developed testbenches for system-level and module-level verification, and executed logic synthesis and timing analysis in Vivado.

SPOTTER — AI Fitness Platform

- ❖ Architected and deployed a full-stack platform to analyze exercise form in real-time from video, integrating machine learning models for pose estimation and movement classification with a responsive infrastructure (**AWS**, **Flask**, **nginx**) to deliver immediate technique feedback to users.

Real-Time Transcription & Translation Learning Overlay

- ❖ Built a desktop app with **Electron**, **Typescript**, **Ollama** and a speech-to-text model to provide real-time translations for foreign media without subtitles.
- ❖ Used **WASAPI** to enable app to operate on audio across websites, web browsers, and video games.
- ❖ Processed transcribed and translated text to color-code words with associated meanings, to assist with language learning.

Mini-Amazon Marketplace Website

- ❖ Developed a CRUD-compliant e-commerce application using **Python (Flask)** and **PostgreSQL**, simulating complex user-product-order lifecycles.
- ❖ Designed a normalized relational database schema and optimized SQL queries to manage transaction history and inventory tracking.
- ❖ Implemented RESTful API endpoints and dynamic front-end routing to ensure a seamless, state-persistent user experience.

Automated Crop Protection System

- ❖ Engineered a distributed IoT system on **ESP32 and Raspberry Pi**, utilizing a **YOLO**-based computer vision pipeline for autonomous pest identification.
- ❖ Optimized video processing using **FFmpeg** for frame capture and preprocessing, ensuring high accuracy under strict hardware and power constraints.
- ❖ Implemented WiFi-based inter-subsystem communication to synchronize sensing and deterrence logic in real-time.
- ❖ Achieved continuous, successful operation even under harsh environmental conditions (rain, snow, and lack of sunlight for solar power).

Network Router, Client and Server

- ❖ Designed to accommodate HTTP requests, ping, trace route algorithms, & network protocols: Ethernet, IP, UDP, TCP, ICMP, RIP, ARP. Written in **C**.

Traceable Legal Contract Analysis

- ❖ Used a Semantic Role Labeling model to build a graph of the relationships and obligations in a dataset of real-world contracts (**PyTorch**).

EXTRACURRICULARS AND LEADERSHIP

- ❖ Member of Quad Council Exec Team. Coordinated community events (barbeques, on-campus farmer's markets, and networking events).
- ❖ Community Lead and Head Chef at Duke Smart Home, a live-in lab centered on IoT and sustainability. Organized and cooked for weekly house dinners for 20+ residents and guests.
- ❖ Self taught guitarist of 5 years; self taught pianist of 12 years.
- ❖ Speaker of intermediate Spanish and beginner's Mandarin.

SOFTWARE AND TECHNICAL SKILLS

- ❖ **Programming Languages:** Java, Python, Typescript, C, C++, C#, HTML/CSS, Assembly, Verilog, SQL (PostgreSQL).
- ❖ **Frameworks:** Flutter, LangGraph, Swing, Flask, Django.
- ❖ **Dev Tools / OS:** Git, Linux, FreeRTOS, Armbian, Windows, LaTeX, Vivado, AWS, IBM Cloud, Firebase, LLMs, AI IDEs (Cursor, Claude Code).
- ❖ **Hardware:** Embedded Systems, FPGA, ESP32, Arduino, CAD (Fusion 360), Logisim, Ultimaker Cura, Meshmixer.
- ❖ **Creative Development:** Adobe (Photoshop, Premiere Pro and After Effects), Unity, SGO Mistika VR, Insta360 Studio.